Project Demo Planning

Date	30 June 2026
Team ID	
Project Name	Rising water
Maximum Marks	1 Mark

Project Demo Planning:
A well-structured demo plan ensures that the team presents the project effectively, covering all key aspects in a clear and organized manner. This document outlines the plan for demonstrating the project, including the flow of the demo, key features to highlight, and responsibilities of each team member.

S.No	Demo Section	Description	Duration (mins)	Responsible Member
1	Introduction & Problem Statement	Present the flood prediction problem and the need for an ML-based early warning system.	0.5	Pandiri Gireesh
2	Solution Architecture	Explain the three-tier architecture: Flask backend, XGBoost ML engine, and PostgreSQL database.	0.5	Pandiri Gireesh
3	ML Model Training Pipeline	Walk through train.py: data preprocessing, SMOTE balancing, model comparison, and XGBoost selection.	1	Pandiri Gireesh
4	Live Feature Demonstration	Demonstrate User Registration, Login, Prediction (/predict), History Dashboard, and Reports page live on localhost.	1.5	Pandiri Gireesh
5	Performance & Load Testing Results	Show Locust load test results proving 53 Req/sec throughput with 0% error rate.	0.5	Pandiri Gireesh

Demo Flow Summary:

Step	Activity	Notes
1	Introduction & Problem Statement	Briefly explain how floods cause devastation and why early prediction is critical.
2	Solution Overview	Highlight XGBoost model accuracy (96%+ ROC-AUC), Flask routing, and PostgreSQL integration.
3	Live Feature Demonstration	Open http://localhost:5000, register a user, make a prediction, and view history dashboard live.
4	Q&A Session	Be prepared to explain SMOTE, StandardScaler, connection pooling, and Flask-Caching decisions.